

Package ‘BirdPipe’

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Type Package

Title R-package for managing the workflow with the BirdPipe recording device

Version 0.1.0

Description BirdPipe is an open-source recording device for automated data collection and classification of birds, bats and other vocal animals. This R-package helps managing the BirdPipe workflow.

Imports jsonlite, readxl, sonicscrewdriver, terra, tuneR, sf, xlsx

License GPL-3

Encoding UTF-8

LazyData true

RoxygenNote 7.3.3

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Contents

extract_community_matrix	1
extract_validated_community_matrix	2
highest_classification_per_segment	3
load_data	3
play_sound	4
plot_detections	5
plot_recordings	5
Index	7

extract_community_matrix
<i>extract_community_matrix</i>

Description

Extracts a community matrix from BirdPipe data

Usage

```
extract_community_matrix(
  BirdPipe.data = BirdPipe.data,
  threshold = 0,
  name.to.use = "species_name"
)
```

Arguments

<code>BirdPipe.data</code>	A BirdPipe data object
<code>threshold</code>	Classification threshold to be used to filter the data (influences number of detected species)
<code>name.to.use</code>	Which names to use for the species

Value

A community matrix, i.e. samples (=segments) times species matrix

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
extract_community_matrix(BirdPipe.data = BirdPipe.data);
```

```
extract_validated_community_matrix
      extract_validated_community_matrix
```

Description

Extracts a validated community matrix from BirdPipe data and manual validation file

Usage

```
extract_validated_community_matrix(
  BirdPipe.data = BirdPipe.data,
  manual.validation.file = NULL,
  validated = c("Y")
)
```

Arguments

<code>BirdPipe.data</code>	A BirdPipe data object
<code>manual.validation.file</code>	Excel file of manual validation results
<code>validated</code>	Which strings are considered as validated detections

Value

A community matrix, i.e. samples (=segments) times species matrix

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data", tgre);
extract_validated_community_matrix(BirdPipe.data = BirdPipe.data, manual.validation.file = "detections_validated.xlsx")
```

highest_classification_per_segment	
	<i>highest_classification_per_segment</i>

Description

Finds the vocalization with highest classification probability for each species and segment

Usage

```
highest_classification_per_segment(
  BirdPipe.data,
  threshold = 0.5,
  manual.validation.file = NULL
)
```

Arguments

BirdPipe.data	A BirdPipe data object
threshold	Classification threshold to be used to filter the data
manual.validation.file	Excel file to be written for entering manual validation results

Value

A named list where the locations of highest classifications are given for each species

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
highest_classification_per_segment(BirdPipe.data = BirdPipe.data);
```

load_data	<i>load_data</i>
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Description

Loads BirdPipe geojson data: recordings and species detections

Usage

```
load_data(BirdPipe.folder, threshold = 0, geojson.folder = "geojson")
```

Arguments

- BirdPipe.folder The name of the BirdPipe folder
- threshold Classification threshold to be used to filter the data
- geojson.folder The name of the geojson folder within the BirdPipe data folder

Value

- A named list with elements \$recordings and \$detections
- A BirdPipe data object

Examples

```
load_data(BirdPipe.folder = "BirdPipe_demo_data");
```

play_sound	<i>play_sound</i>
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Description

Plays part of audio file (.flac or .wav) e.g. for manual validation purpose

Usage

```
play_sound(  
  BirdPipe.folder,  
  audio_folder = NULL,  
  audio_filename,  
  offset,  
  classification.interval = 3,  
  buffer = 1  
)
```

Arguments

- BirdPipe.folder The BirdPipe data folder
- audio_folder The name of the audio folder within the BirdPipe data folder
- audio_filename The name of the audio file
- offset The beginning of the 3 second segment from which the vocalization is classified
- classification.interval Length of the classification interval in seconds
- buffer Number of seconds added before and after the classification interval

Examples

```
play_sound(BirdPipe.folder = "BirdPipe_demo_data", audio_filename = "field-4_2025-12-22T10_05_03.011300Z.wav")  
#This function call plays a vocalization of Eurasian Bullfinch from the R-package demonstration data.  
#Note that Mac users may obtain a "Permission denied" error. If this happens, you may need to set the folder where
```

plot_detections	<i>plot_detections</i>
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Description

Plots BirdPipe detections as timeline

Usage

```
plot_detections(  
  BirdPipe.data,  
  name.to.use = "species_name",  
  species.order = "alphabetical",  
  jitter = 0,  
  threshold = 0  
)
```

Arguments

BirdPipe.data	A BirdPipe data object
name.to.use	Which names to use for the species
species.order	In which order to show the species ("alphabetical" or "number_of_detections")
jitter	the amount of random jitter added to shows overlapping datapoints
threshold	Classification threshold to be used to filter the data (influences number of detected species)

Examples

```
BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");  
plot_detections(BirdPipe.data = BirdPipe.data)
```

plot_recordings	<i>plot_recordings</i>
-----------------	------------------------

Description

Plots BirdPipe recordings as timeline and on a map

Usage

```
plot_recordings(  
  BirdPipe.data,  
  show.timeline = TRUE,  
  show.map = TRUE,  
  background.map = NULL,  
  show.number.of.species = FALSE,  
  threshold = 0,  
  show.segment.id = FALSE,
```

```

    number.of.species.color = "white",
    segment.id.color = "white",
    jitter = 0,
    spatial.buffer = 1
)

```

Arguments

<code>BirdPipe.data</code>	A BirdPipe data object
<code>show.timeline</code>	Boolean variable (TRUE/FALSE) describing whether timeline is to be plotted
<code>show.map</code>	Boolean variable (TRUE/FALSE) describing whether map is to be plotted
<code>background.map</code>	optional terra/SpatRaster object to be plotted as background map
<code>show.number.of.species</code>	Boolean variable (TRUE/FALSE) describing whether number of detected species is shown
<code>threshold</code>	Classification threshold to be used to filter the data (influences number of detected species)
<code>show.segment.id</code>	Boolean variable (TRUE/FALSE) describing whether segment id:s are shown
<code>number.of.species.color</code>	color for plotting number of species
<code>segment.id.color</code>	color for plotting segment id:s
<code>jitter</code>	the amount of random jitter added to spatial locations to avoid their possible overlap (proportional to data extent)
<code>spatial.buffer</code>	size of margins added to the map (proportional to data extent)

Examples

```

BirdPipe.data = load_data(BirdPipe.folder = "BirdPipe_demo_data");
plot_recordings(BirdPipe.data = BirdPipe.data)

```

Index

`extract_community_matrix`, [1](#)
`extract_validated_community_matrix`, [2](#)
`highest_classification_per_segment`, [3](#)
`load_data`, [3](#)
`play_sound`, [4](#)
`plot_detections`, [5](#)
`plot_recordings`, [5](#)